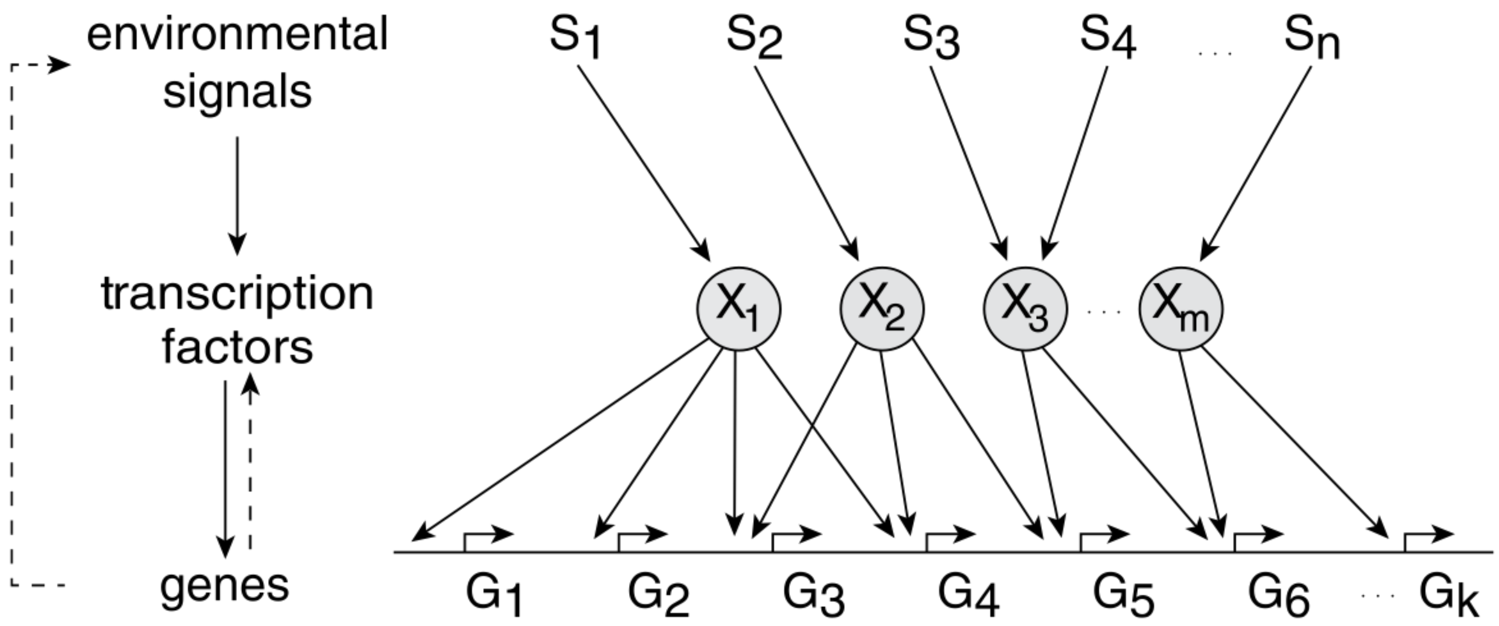
Lecture 1 Chemotaxis

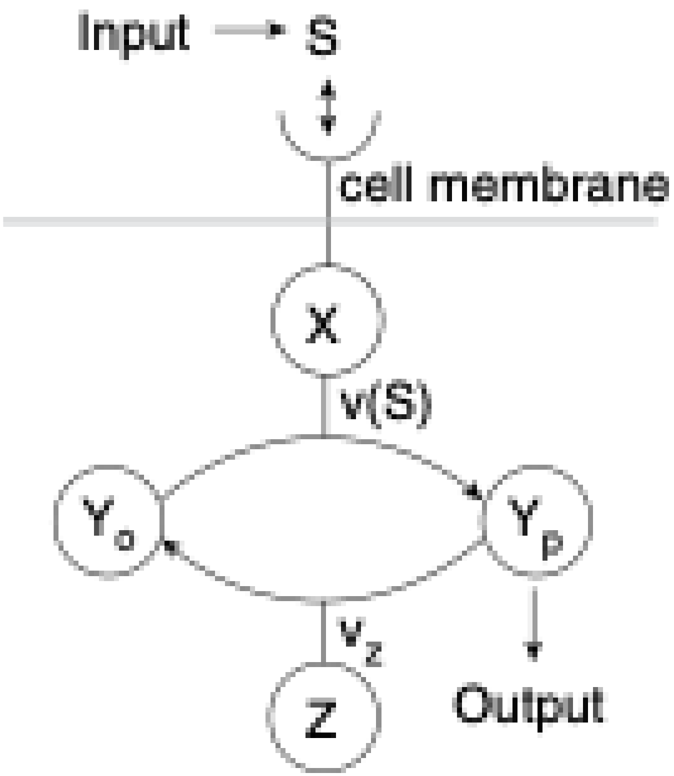
2026/09/01

How do cells think ?



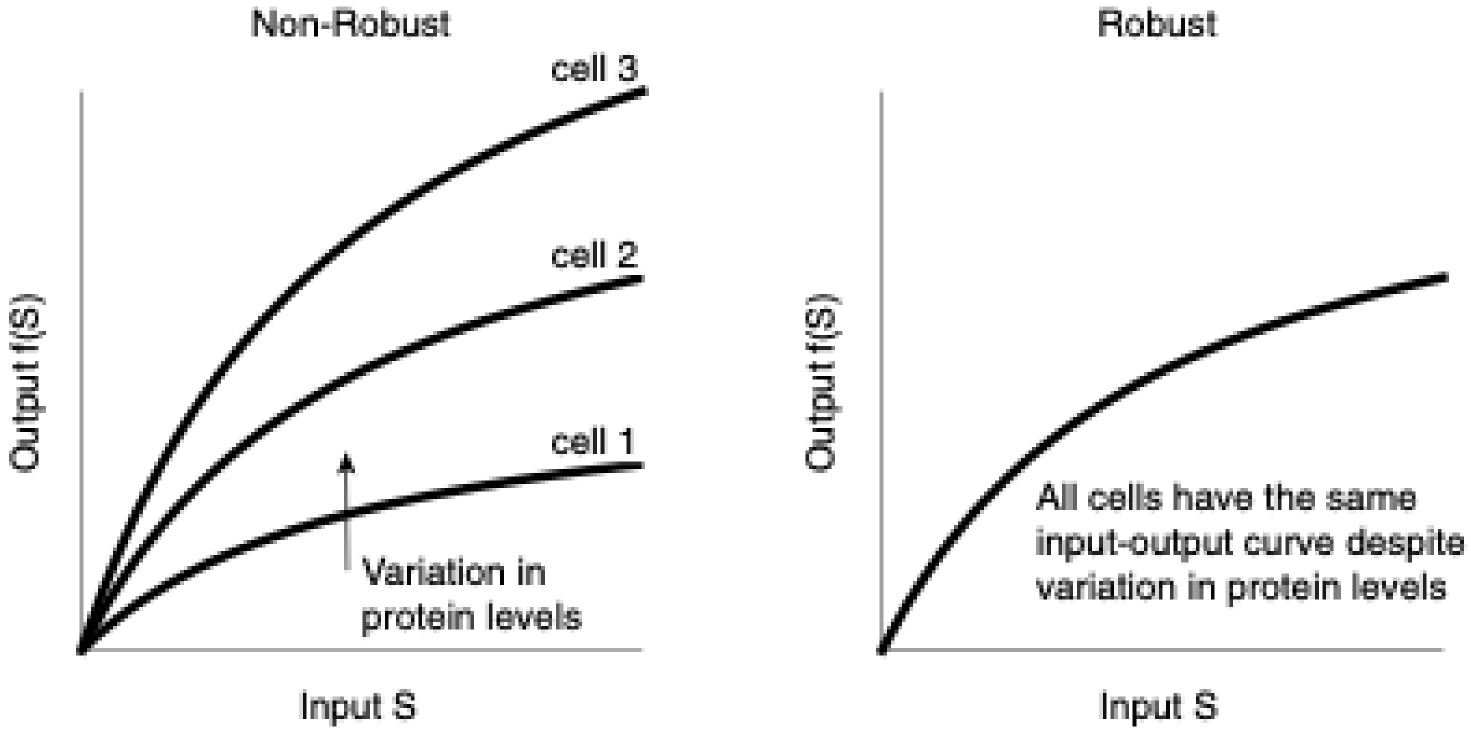
Source: Alon, An Introduction to Systems Biology, 2nd ed. (2020)

Simple models of signal-transduction circuits are not robust



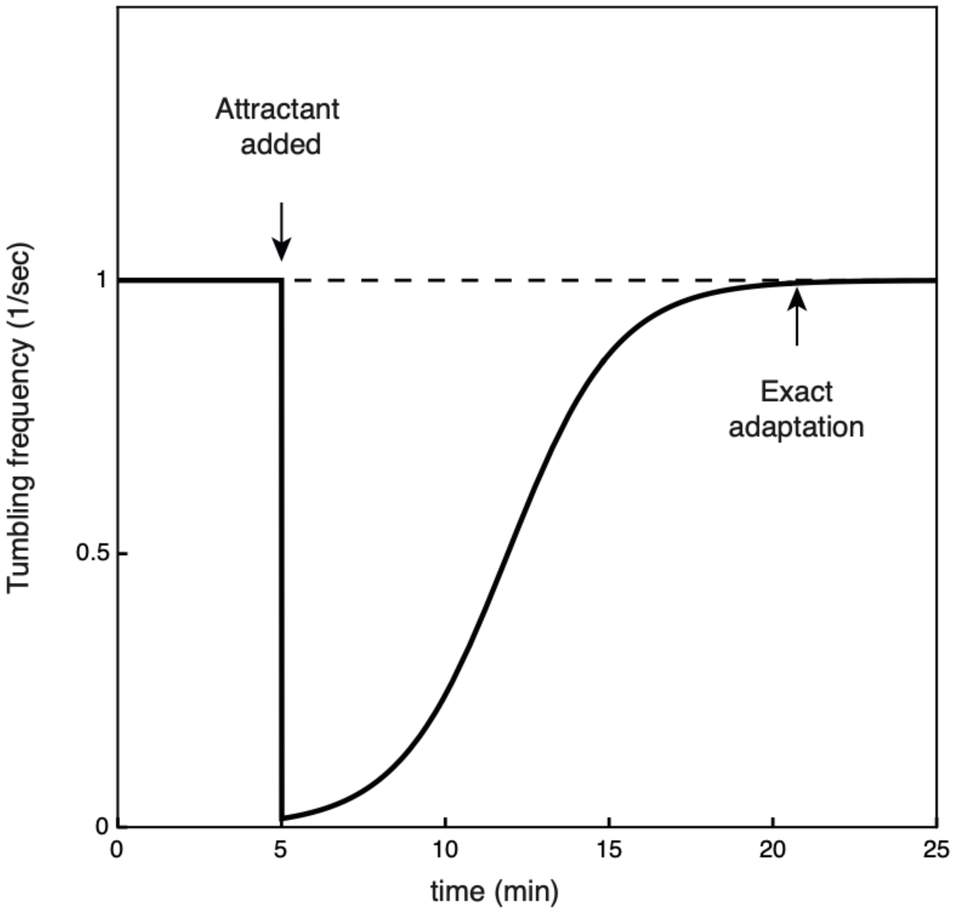
Source: Alon, An Introduction to Systems Biology, 2nd ed. (2020)

Biological circuits must operate robustly in the face of variations



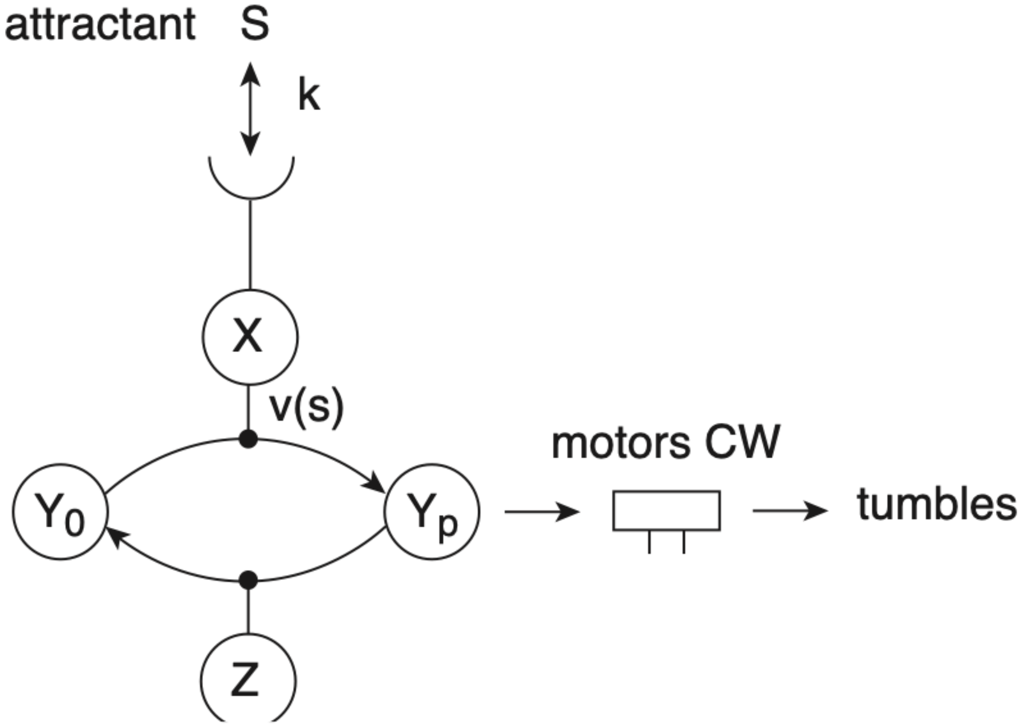
Source: Alon, An Introduction to Systems Biology, 2nd ed. (2020)

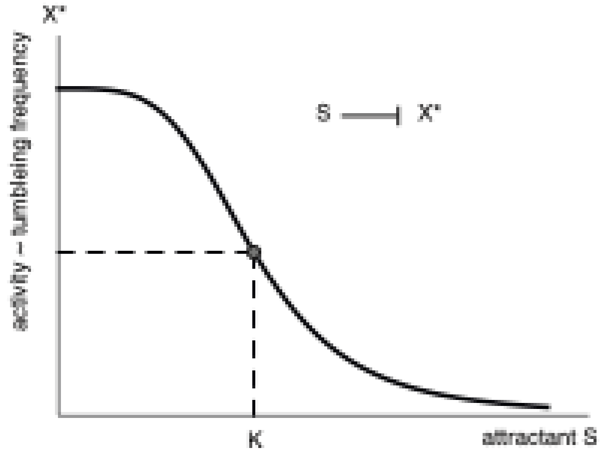
A basic feature of the chemotaxis response can be revealed by a simple experiment



Source: Alon, An Introduction to Systems Biology, 2nd ed. (2020)

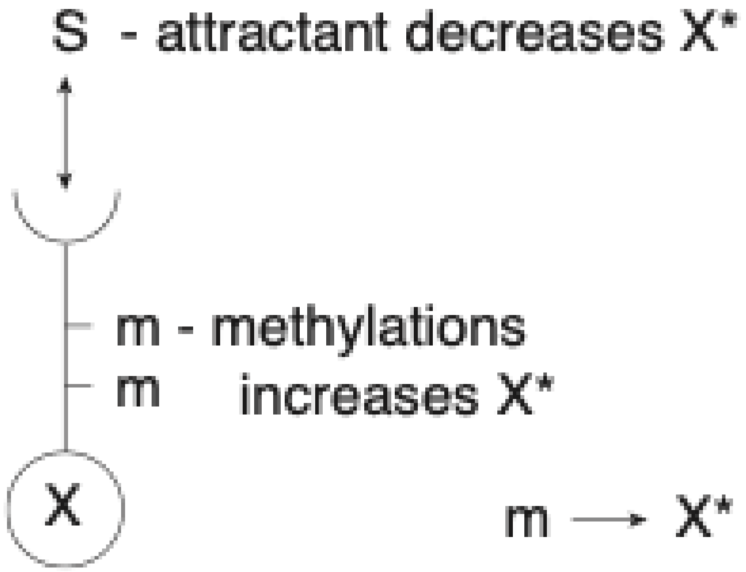
Attractant lowers the activity of X

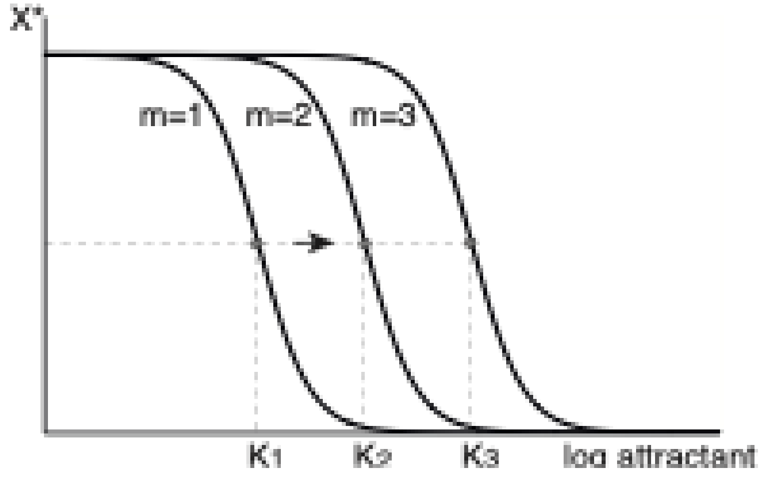




Source: Alon, An Introduction to Systems Biology, 2nd ed. (2020)

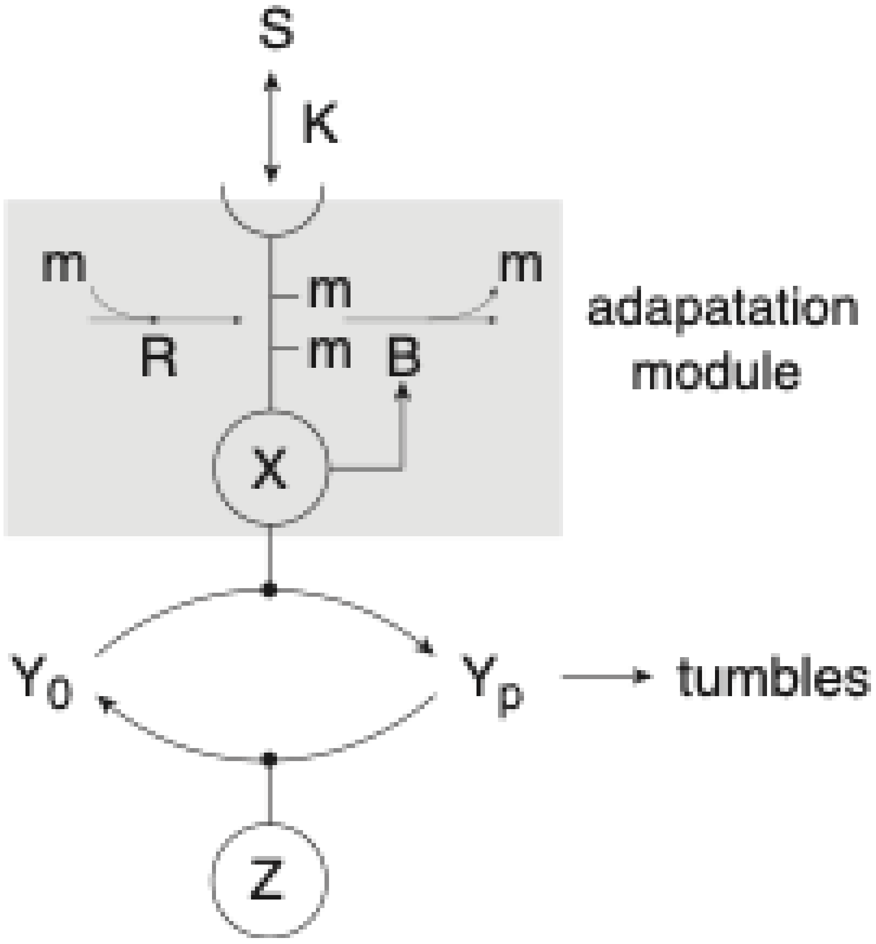
Adaptation is due to slow modification of X that increases its activity





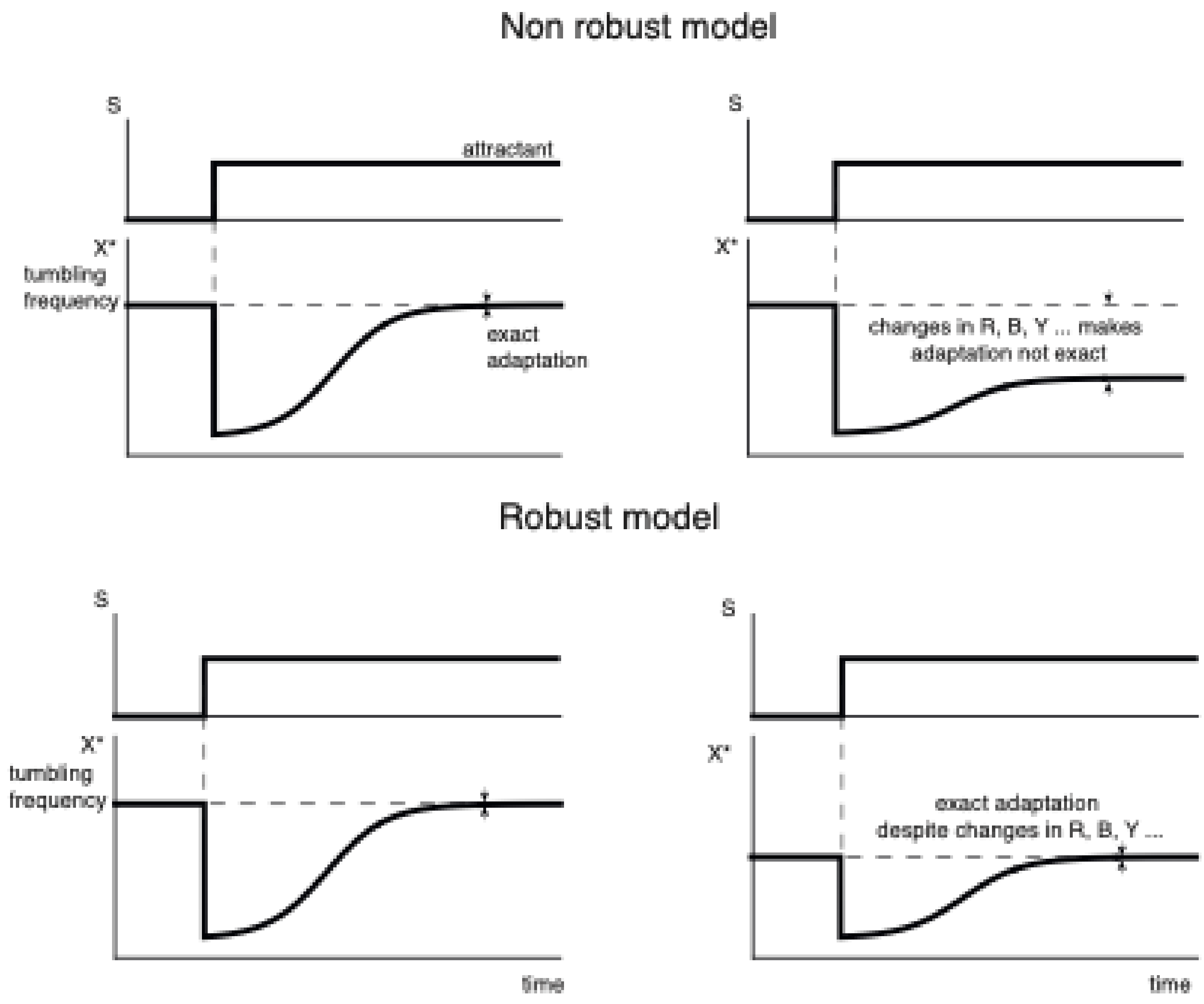
Source: Alon, An Introduction to Systems Biology, 2nd ed. (2020)

CheR and CheB form an adaptation module



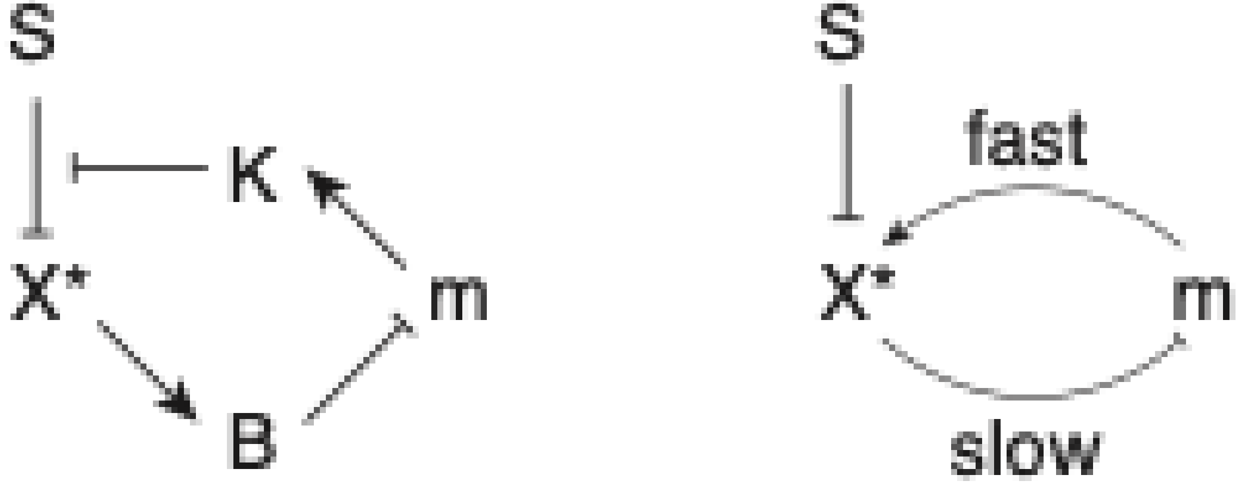
Source: Alon, An Introduction to Systems Biology, 2nd ed. (2020)

Many model adapt – but only some do so robustly



Source: Alon, An Introduction to Systems Biology, 2nd ed. (2020)

Barkai–Leibler's key proposal: B demethylates only active X, forming a slow negative feedback loop



Source: Alon, An Introduction to Systems Biology, 2nd ed. (2020)
Barkai, N., Leibler, S. 1997 Nature

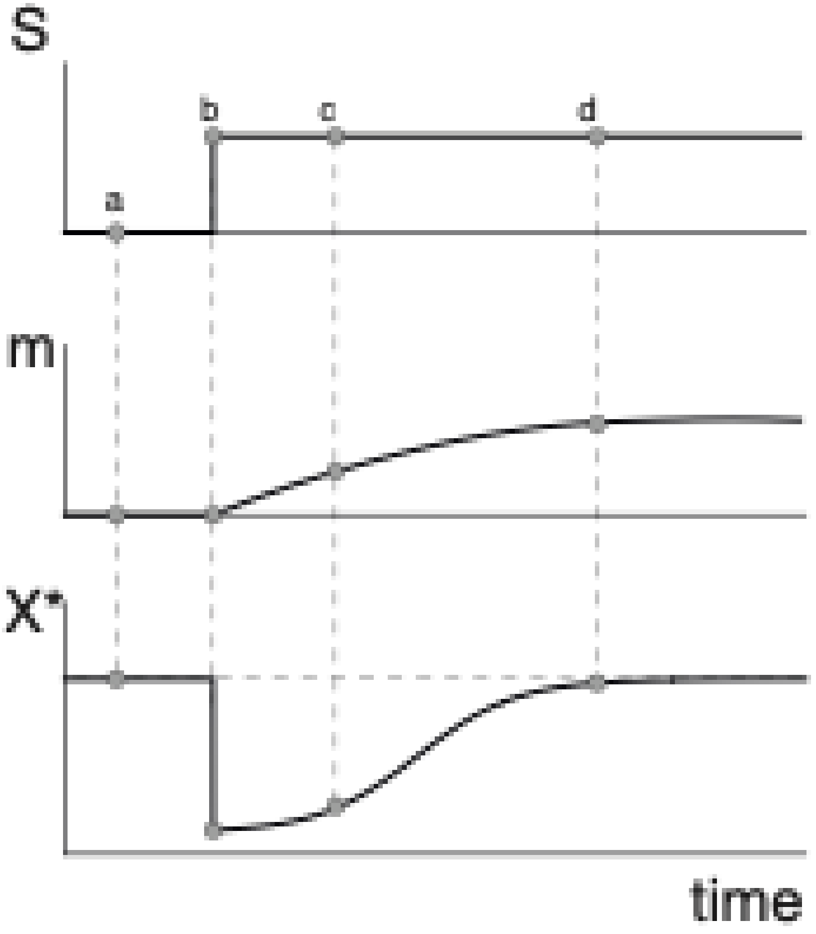
The Barkai-Leibler model explains robust exact adaptation

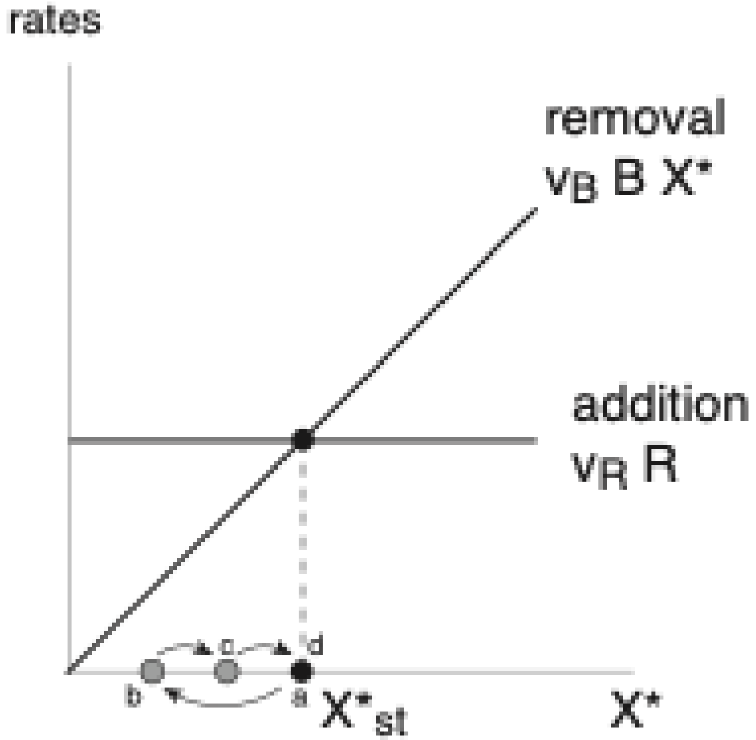
$$dm/dt = Vr\ R - Vb\ B\ X^*$$

$$X^*st = VrR/VbB.$$

Source: Alon, An Introduction to Systems Biology, 2nd ed. (2020)
Barkai, N., Leibler, S. 1997 Nature

Exact adaptation is robust; baseline activity and adaptation time are not





Source: Alon, An Introduction to Systems Biology, 2nd ed. (2020)
U. Alon et al 1999 Nature

Barkai-Leibler model is mapped to integral feedback

$$\frac{dm}{dt} = VbB(X_{st}^{*} - X^{*})$$

$$m(t) \sim \int_{-\infty}^{t} (X_{st}^{*} - X^{*}(t))\, dt \sim \int error(t) dt$$

Source: Alon, An Introduction to Systems Biology, 2nd ed. (2020)
T. Yi et al 2000 PNAS

Alon et al. experimentally test robust exact adaptation

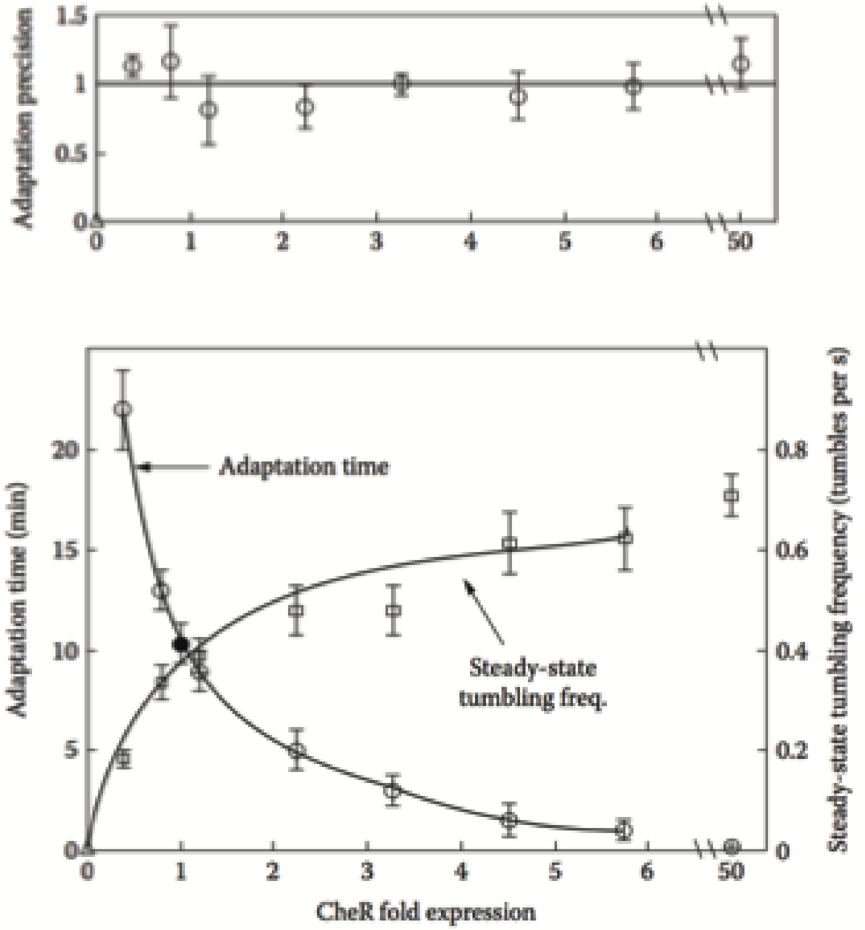
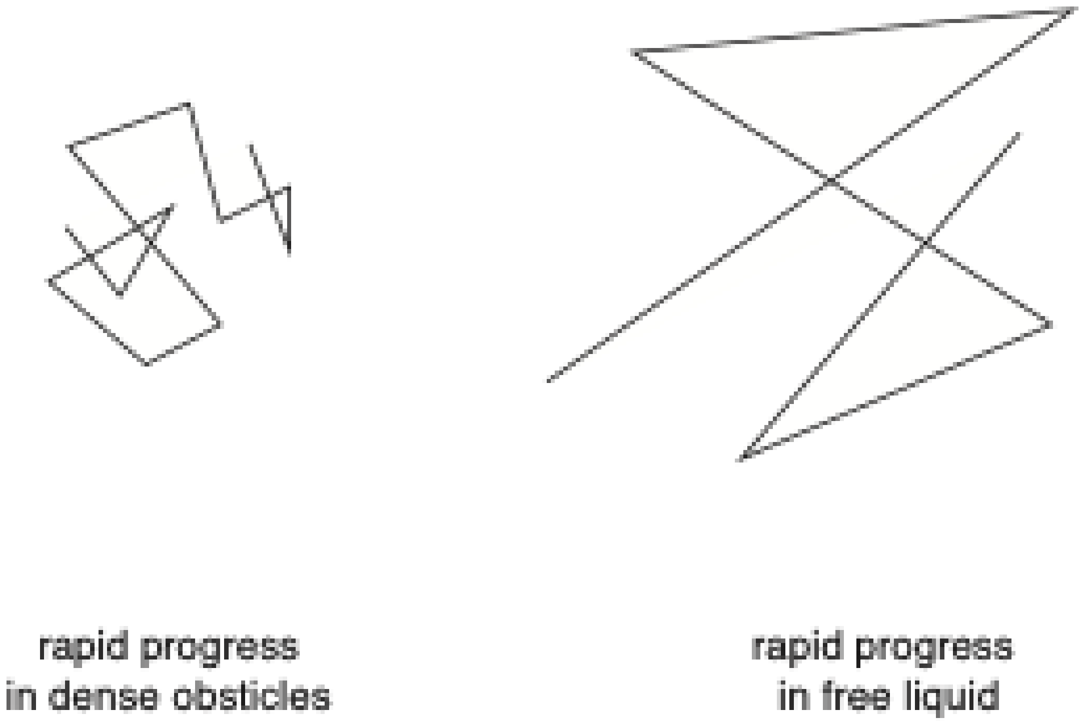


Table 1 Effect of variation in chemotaxis protein levels on behaviour

Protein varied	Fold expression	Strain background	Steady-state tumbling frequency (s^{-1})	Adaptation time (min)	Precision of adaptation
Wild type	1.0	Wild type (RP437)	0.44 ± 0.03	10 ± 1	0.98 ± 0.05
CheB*	0.4 ± 0.1	ΔcheB (RP4972)	0.66 ± 0.05	7 ± 1	0.98 ± 0.12
CheB*	12 ± 3	ΔcheB (RP4972)	0.14 ± 0.02	15 ± 1	1.09 ± 0.11
CheBc†	~1‖	ΔcheB (RP4972)	0.74 ± 0.06	9 ± 2	0.90 ± 0.13
CheY‡	0.2 ± 0.1	ΔcheY,Z (RP5231)	0.24 ± 0.04	11 ± 3	1.04 ± 0.08
CheZ	0	ΔcheZ (RP1616)	1.6 ± 0.1	10 ± 2	1.1 ± 0.14
Tar, Tap, CheR,B,Y,Z§	5 ± 2¶	Wild-type (RP437)	0.30 ± 0.06	3 ± 1	1.04 ± 0.07

Source: Alon, An Introduction to Systems Biology, 2nd ed. (2020)
U. Alon et al 1999 Nature

Protein concentration fluctuations result in individuality in chemotaxis



Source: Alon, An Introduction to Systems Biology, 2nd ed. (2020)